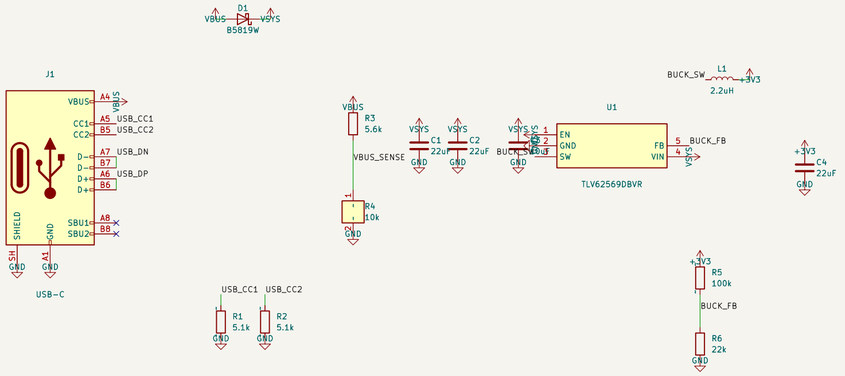
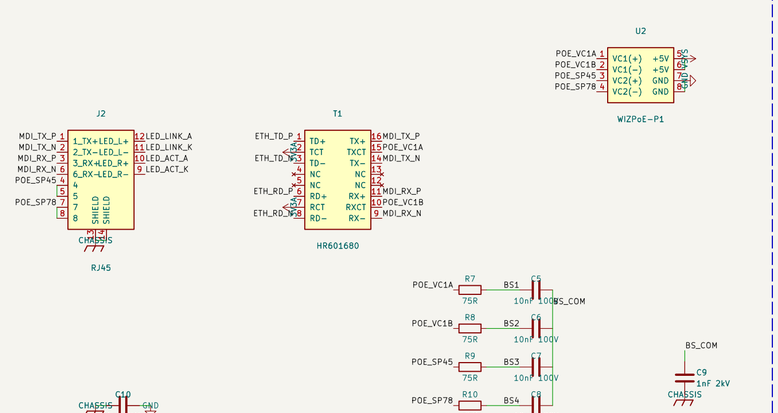
USB-C, VSYS and 3V3 buck



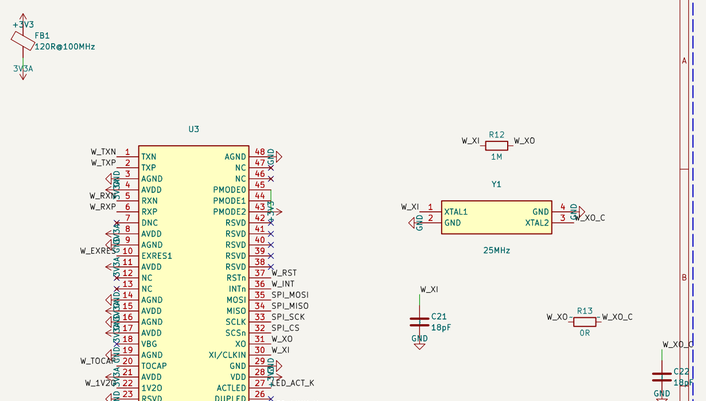
USB VBUS and the PoE module's 5 V meet on VSYS: D1 keeps PoE out of USB.
TLV62569: VOUT = 0.6 V × (1 + R5/R6) = 3.33 V; 2.2 uH + 22 uF (T1 table 4).

PoE: RJ45, magnetics, WIZPoE-P1 (isolated 802.3af)



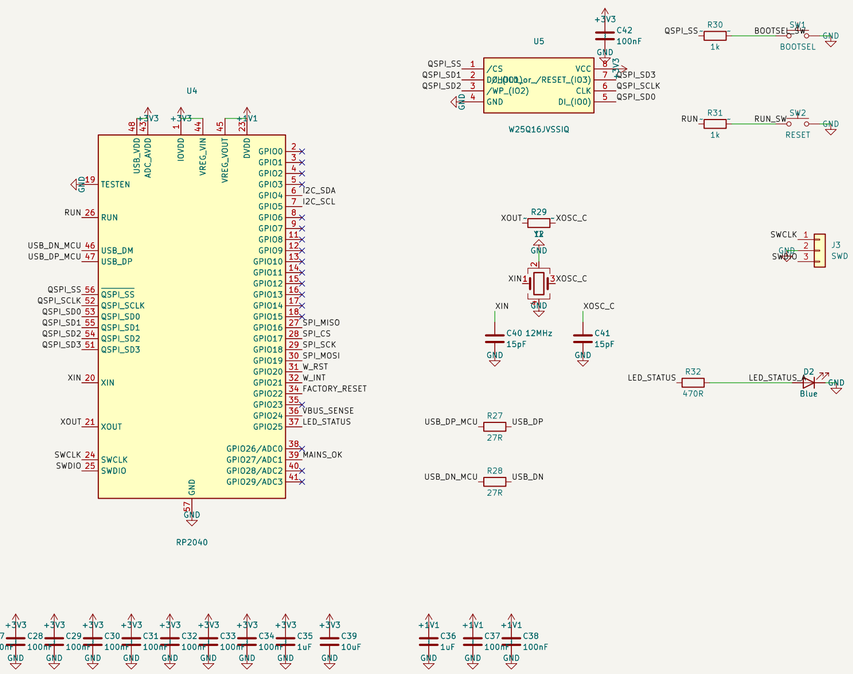
Mode A: T1 line centre taps → VC1. Mode B: spare pairs 4/5, 7/8 → VC2; U2 rectifies both.
Bob Smith: 75R + 10 nF/100 V per pair, 1 nF/2 kV to chassis. W5500 has no auto-MDIX: MDI.
MDI side (J2, T1 pins 9–16, POE_*, BS*) keeps 2 mm from logic copper.

W5500 Ethernet (SPI0: GP16–GP21)



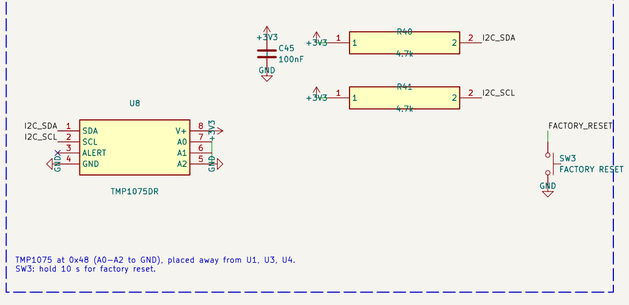
PMODE 111: auto-negotiation. Pin 23 RSVD to GND; TOCAP 4.7 uF; 1V2O 10 nF (datasheet v1.1.0).
TX: 3.3R series, 49.9R to the 3V3A centre tap, RX: 6.8 nF coupling, 49.9R to C25.

RP2040, flash, clock, USB, buttons, SWD



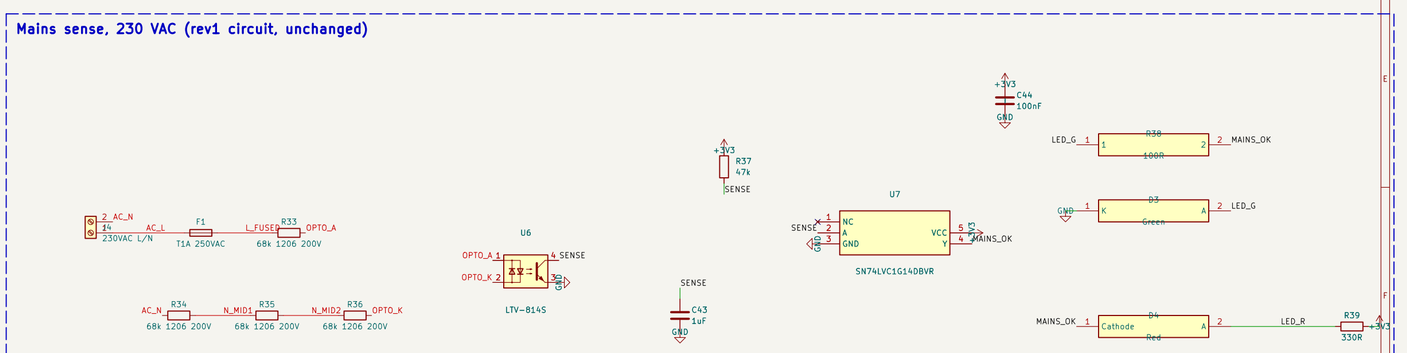
Raspberry Pi Pico core: 12 MHz crystal, W25Q16 2 MB flash, BOOTSEL on QSPI_SS.
Pin map unchanged (firmware board = pico): GP4/5 I2C0, GP16–21 W5500, GP22 factory reset,
GP24 VBUS sense, GP25 status LED, GP27 MAINS_OK.

Ambient temperature, factory reset



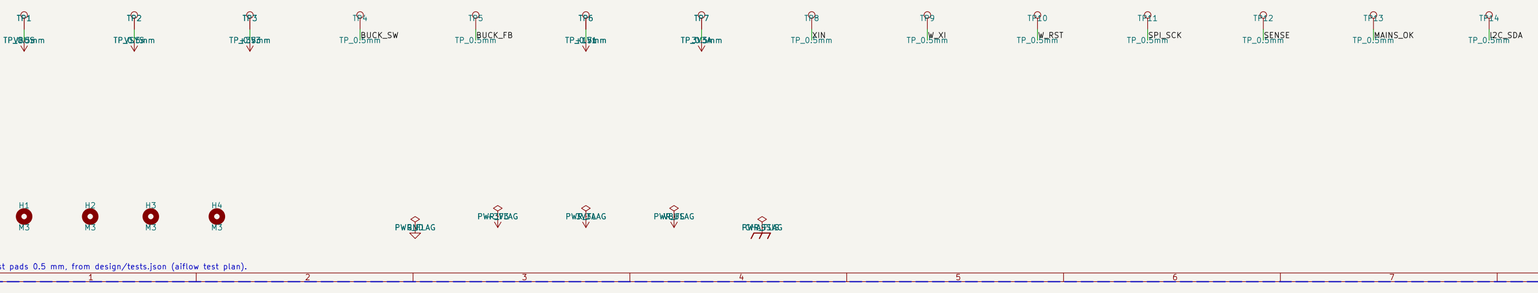
TMP1075 at 0x48 (A0–A2 to GND), placed away from U1, U3, U4.
SW3: hold 10 s for factory reset.

Mains sense, 230 VAC (rev1 circuit, unchanged)



DANGER 230 VAC: J4, F1, R33–R36, U6 inputs: 6.5 mm clearance, 6.4 mm creepage to everything else.
U6 pulls SENSE low on both half cycles; C43/R37 bridge the zero crossings; U7 gives MAINS_OK (HIGH = mains).

Test pads, power flags, mounting holes



Test pads 0.5 mm, from design/tests.json (aiflow test plan).

Generated by kicad/tools/gen_sch.py from design/netlist.json: edit those, not this file
RP2040 + W5500 + WIZPoE-P1 on one board; mains detect GP27, TMP1075, factory reset GP22
Kenneth Foli Jørgensen

Sheet: /
File: rf-power-monitor-rev2.kicad_sch

Title: RF-Power-Monitor rev2

Size: A2	Date: 2026-11-02	**Rev: 2.0**
KiCad E.D.A. 10.0.7		Id: 1/1